

Mechanism and Efficacy of L-Ornithine L-Aspartate (LOLA) in Chronic Liver Disease Management

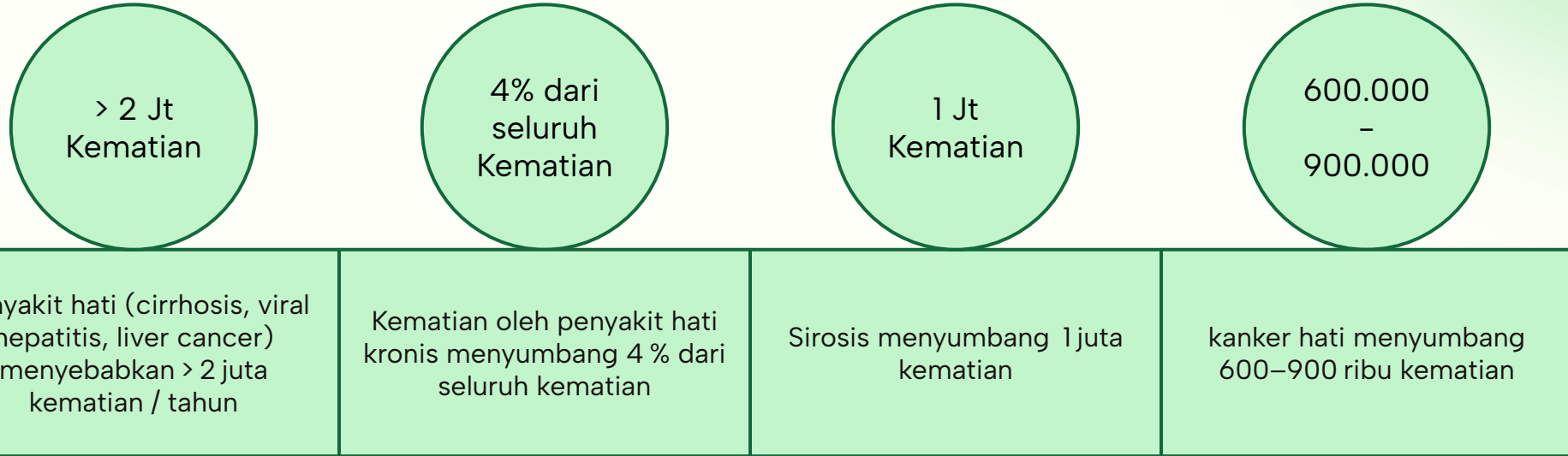
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01

Chronic Liver Diseases

Epidemiologi - Global



Mortalitas lebih tinggi di negara berpendapatan rendah-menengah & usia > 65 tahun

Etiologi Utama Penyakit Hati Kronik

30,1%

MASLD/NAFLD kini penyebab
CLD paling sering
Prevalensi global 30,1 %

↑↑
50%

Prevalensi MASLD meningkat
50 % (1990–2019)

70%
& 75%

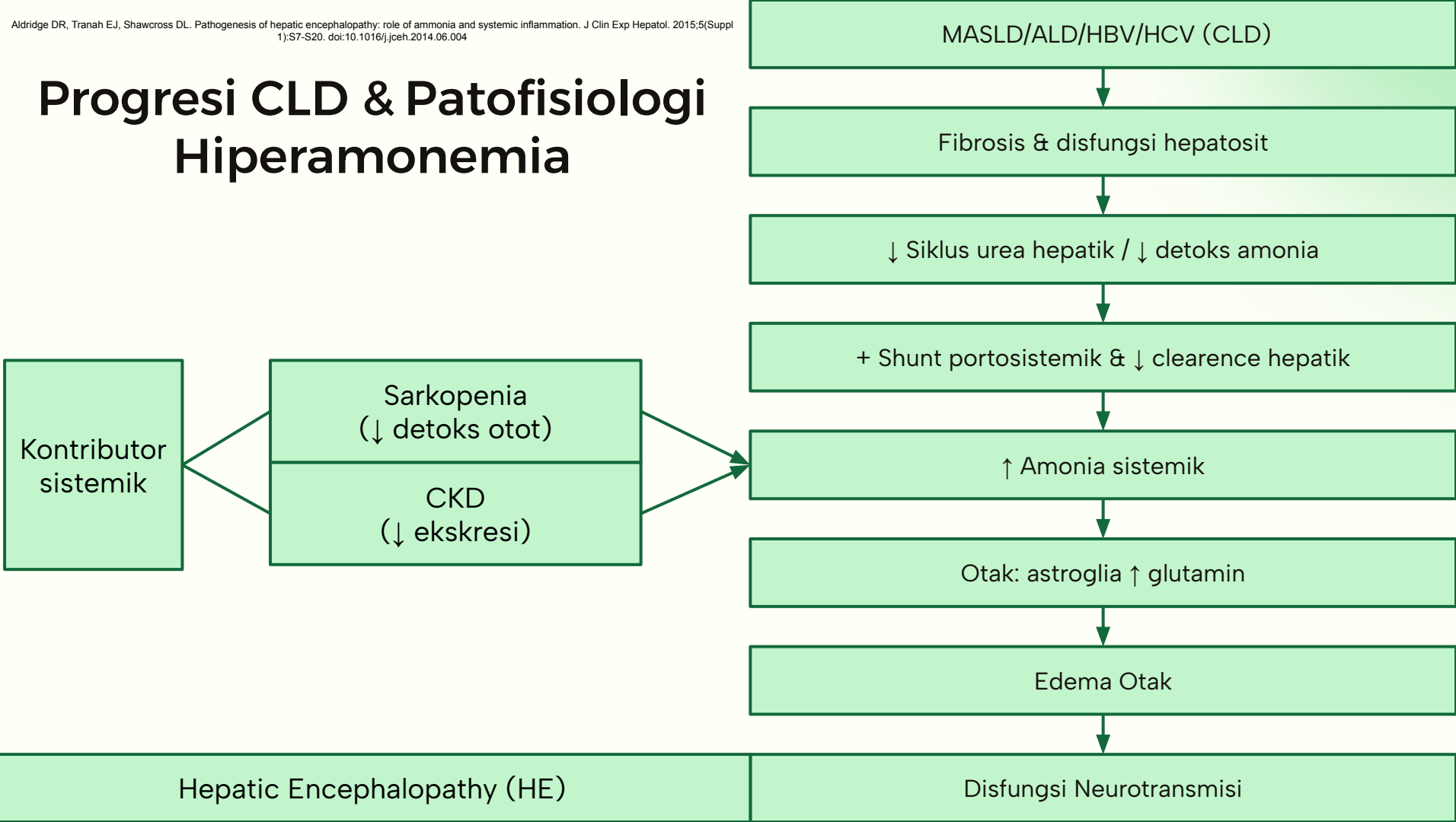
MASLD terjadi pada 70 %
orang overweight dan 75 %
penderita obesitas

40%

Prevalensi MASLD pada
populasi non-obesitas dapat
mencapai 40 %

Hepatitis B dan C masih tetap masalah di wilayah tertentu

Progresi CLD & Patofisiologi Hiperamonemia



Epidemiologi Ensefalopati Hepatik (HE)

40%

HE terjadi pada hingga 40 %
penderita sirosis

Survival
1-2 tahun

Median survival setelah
diagnosis HE $\approx$ 2 tahun
hanya 1 tahun pada usia > 65

20-37%

Prevalensi minimal HE (CHE)
20-37 % berdasarkan PHES

11,6/100

Insidensi OHE 11,6 per 100
pasien-tahun (Medicare)

10 % Child A & 25 % Child B mengalami OHE dalam 1 tahun

Faktor Risiko HE pada CLD

38%

Diagnosis CHE sering terlambat
38 % klinisi tidak melakukan pemeriksaan

2,39/100

NAFLD meningkatkan risiko HE
Insidensi 2,39 per 100 pasien-tahun pada fibrosis F4

86%

Diabetes meningkatkan risiko HE 86 % melalui neuropati, konstipasi dan gangguan ginjal

↑1,1%

Usia lanjut memperparah HE
Prevalensi meningkat 1,1%/tahun pada usia > 65

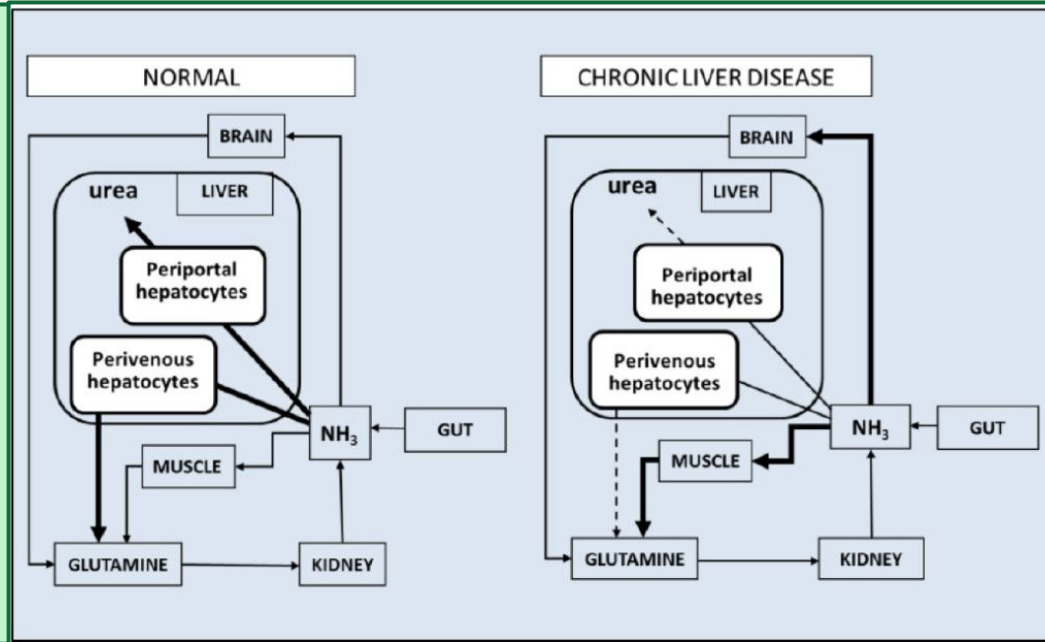
Obesitas & sarcopenia melemahkan metabolisme amonia & memperburuk HE

Amonia in Chronic Liver Disease

Normal:

Periportal hepatosit:
detoks amonia melalui
siklus urea

Perivenous hepatosit:
glutamine synthetase
membersihkan sisa
amonia



CLD:

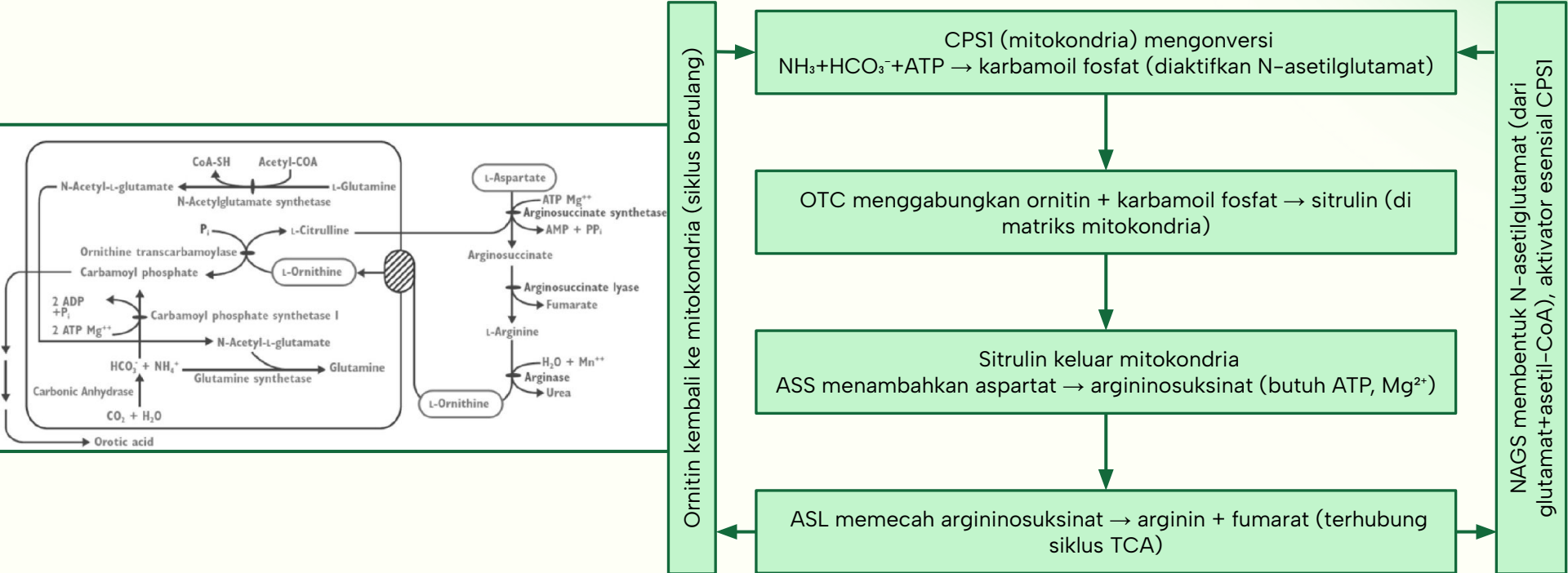
hipertensi portal dan
shunt portosistemik
kaya amonia.

Hilang hingga 85%
hepatosit
periportal/perivenous;
detoks amonia
terganggu

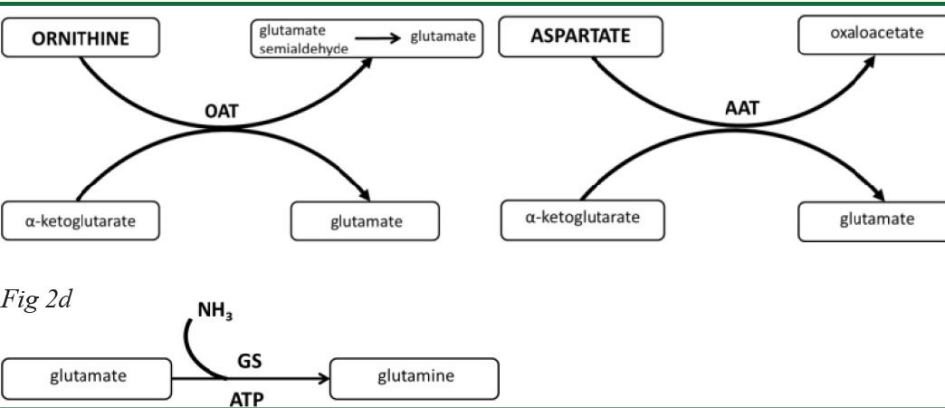
Amonia beredar pada
otak ↑↑
Peningkatan glutamin
otak berkorelasi
derajat ensefalopati
hepatik

Pada sirosis, otot
menggantikan hati
sebagai penghapus
amonia

Siklus Urea



Transaminasi



Ornitin dan aspartat ditransaminasi menghasilkan glutamat

Glutamat adalah substrat wajib enzim glutamine synthetase (GS)

GS terdapat di perivenous hepatocytes, otot, dan otak

↑↑ aliran melalui GS → ↑↑ produksi glutamin pada sirosis

↑↑ glutamin otak → ↑↑ derajat HE

Factor 1: Underlying disease

Type A	Acute liver failure
Type B	Portosystemic/shunt
Type C	Cirrhosis

Factor 2: Severity of manifestation

	Severity manifestation	Clinical description
Covert	Minimal	No clinical manifestations. Abnormal psychometric test. Abnormal neuropsychological test
	Grade I	Personality changes. Attention deficit. Irritability. Addition/Subtraction impairment. Altered sleep rhythm
Overt	Grade II	Lethargy or apathy. Disorientation for time. Obvious personality changes. Inappropriate behavior. Dyspraxia. Asterix
	Grade III	Somnolence to semi-stupor. Responsive to stimuli. Confused. Gross disorientation. Bizarre behavior
	Grade IV	Stupor and coma

Factor 3: Time course

Episodic	Bouts occurring more than 6 mo apart
Recurrent	Time between HE episode ≤ 6 mo
Persistent	Behavior changes always present superimposed with HE exacerbations

Factor 4: Precipitating factors

Infection
GI bleed/constipation diuretic overdose
Renal failure/hypovolemia/hypoxia/hypoglycemia hypokalemia/metabolic alkalosis
Sedative/tranquillizer use

Klasifikasi (AASLD & EASL)

Penyakit yang mendasari
Type A resulting from ALF
Type B resulting predominantly from portosystemic bypass or shunting
Type C resulting from cirrhosis

Perjalanan Waktu
Episodic
Recurrent
Persistent

Tingkat keparahan manifestasi klinis
CHE → Minimal, Grade I
OHE → Grade II, III, IV

Faktor pencetus
Tanpa Faktor pencetus
Dengan Faktor pencetus

Episodic

Infections*
GI bleeding
Diuretic overdose
Electrolyte disorder
Constipation
Unidentified

Recurrent

Electrolyte disorder
Infections
Unidentified
Constipation
Diuretic overdose
GI bleeding

WHC Including MHE	ISHEN	Description
Unimpaired		No encephalopathy at all, no history of HE
Minimal	Covert	Psychometric or neuropsychological alterations of tests exploring psychomotor speed/executive functions or neurophysiological alterations without clinical evidence of mental change
Grade I		• Trivial lack of awareness • Euphoria or anxiety • Shortened attention span • Impairment of addition or subtraction • Altered sleep rhythm
Grade II	Overt	• Lethargy or apathy • Disorientation for time • Obvious personality change • Inappropriate behavior • Dyspraxia • Asterixis
Grade III		• Somnolence to semistupor • Responsive to stimuli • Confused • Gross disorientation • Bizarre behavior
Grade IV		Coma

Vilstrup H, Amodio P, Bajaj J, et al. Hepatic encephalopathy in chronic liver disease: 2014 practice guideline by the European Association for the Study of the Liver and the American Association for the Study of Liver Diseases. J Hepatol 2014;61:642–659.

Stadium HE & Klinis (AASLD & EASL)

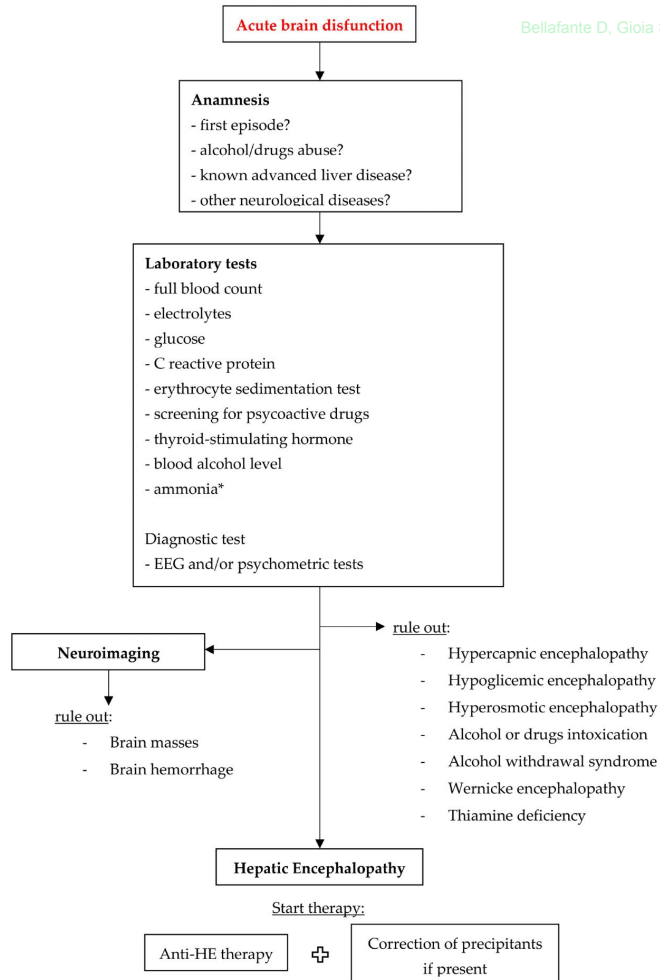
Tingkat keparahan manifestasi klinis

CHE → Minimal, Grade I

OHE → Grade II, III, IV

Diagnosis

1. Anamnesis
2. Lab Test
3. Neuroimaging



*to avoid false positives blood samples should ideally be drawn without applying a tourniquet, placed on ice immediately and quickly sent to the laborator

Manajemen Terapi Farmakologi HE (KASL)

Table 5. Pharmacological options for managing overt hepatic encephalopathy

Non-absorbable disaccharides	Lactulose (20–30 g) should be administered orally 3–4 times per day (an equivalent daily dose of lactitol is 67–100 g). Goals: it should be administered orally until the patient is having at least 2 bowel movements a day. Thereafter, the dose should be titrated to achieve two to three soft stools per day. If patients cannot take medications orally, administration via nasogastric tube might be tried. Enema with lactulose 200 g and 700 mL water might be performed 3–4 times per day in severe cases.
Rifaximin	400 mg three times/day or 550 mg twice/day
Oral BCAA	0.25 g/kg/day
Intravenous LOLA	30 g/day
Albumin	1.5 g/kg/day until clinical improvement or for 10 days, maximum
Polyethylene glycol	A substitute for non-absorbable disaccharides 4 liters orally

BCAA, branched-chain amino acid; LOLA, L-ornithine-L-aspartate.

Terapi HE (AASLD & EASL)

Nonabsorbable Disaccharides

Probiotics

Rifaximin

Neomycin

BCAAs

Other Therapies

L-ornithine L-aspartate (LOLA)

RECOMMENDATION 22

Intravenous (IV) L-ornithine L-aspartate (LOLA) can be used as an alternative or additional agent to treat patients nonresponsive to conventional therapy (GRADE I, B, 2).

02

LOLA

L-Ornithine L-Aspartate (LOLA)

LOLA tersedia sebagai sachet oral 3g dan ampul injeksi 5 g/10 mL

Bioavailabilitas oral tinggi (82 %) dengan puncak plasma 30–60 menit

Infus 5g: $t_{1/2}$ distribusi 15–20 menit
eliminasi 120–150 menit

Dosis oral untuk HE:
9–18g (dibagi menjadi 2–3 dosis)
7–30 hari

Dosis intravena:
20–40 g per hari (infus 4–8 jam)
3–7 hari

Farmakokinetik & Farmakodinamik

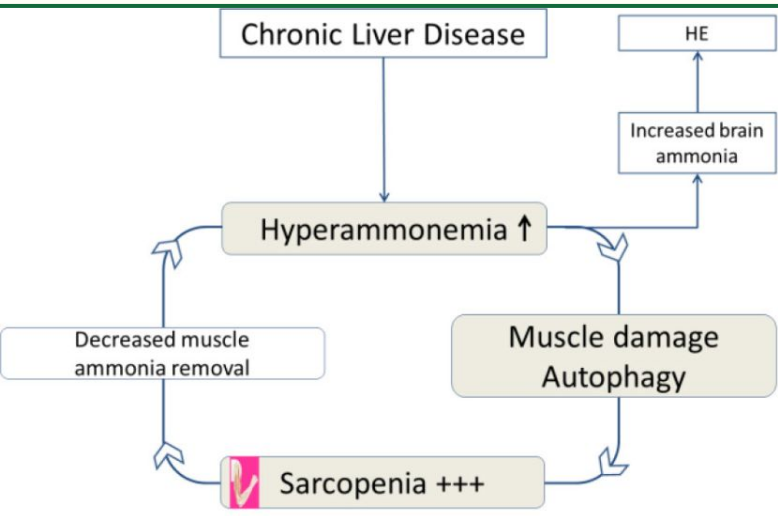
LOLA adalah garam stabil yang terionisasi menjadi L-ornitin dan L-aspartat

L-ornitin didistribusikan ke hati/otot, menstimulasi CPS1 untuk membentuk karbamoil fosfat

L-aspartat meningkatkan aktivitas arginase dan transaminasi menghasilkan glutamat

LOLA tidak terakumulasi; metabolit (urea, glutamin) diekskresikan melalui urine dan otot.

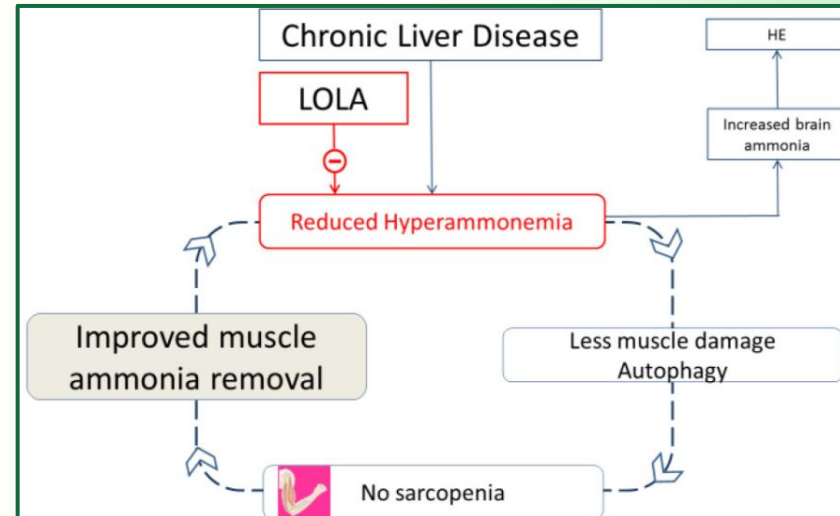
Mekanisme Kerja LOLA



Hiperamonia memicu autophagy dan penurunan sintesis protein otot

Sarkopenia menurunkan ekstraksi amonia → hiperamonia semakin memburuk

Siklus hiperamonia-sarkopenia berulang

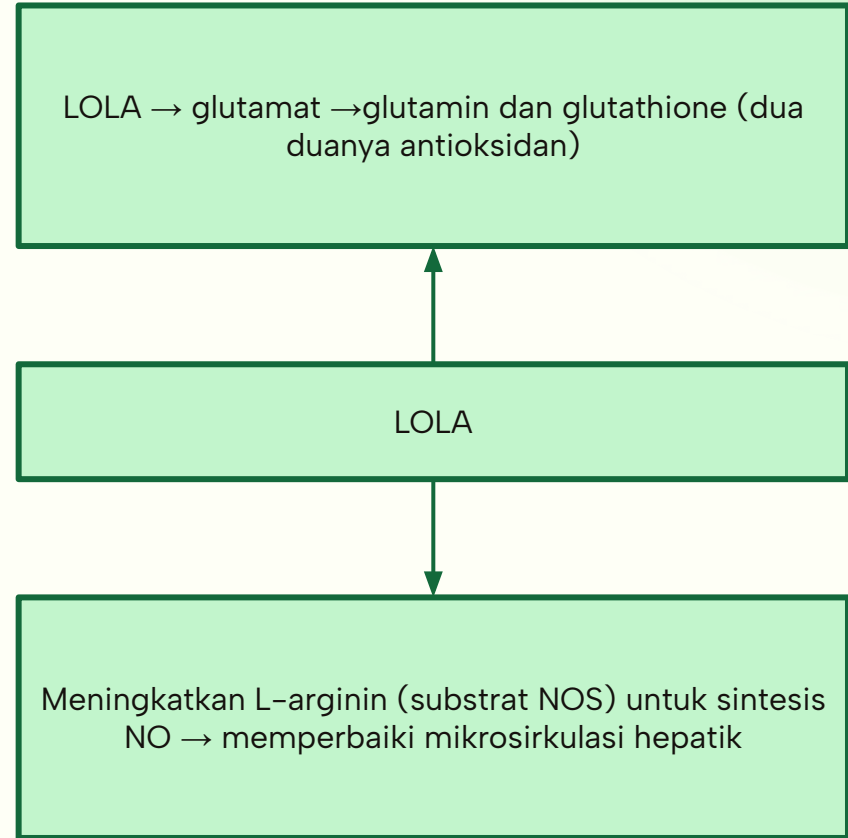
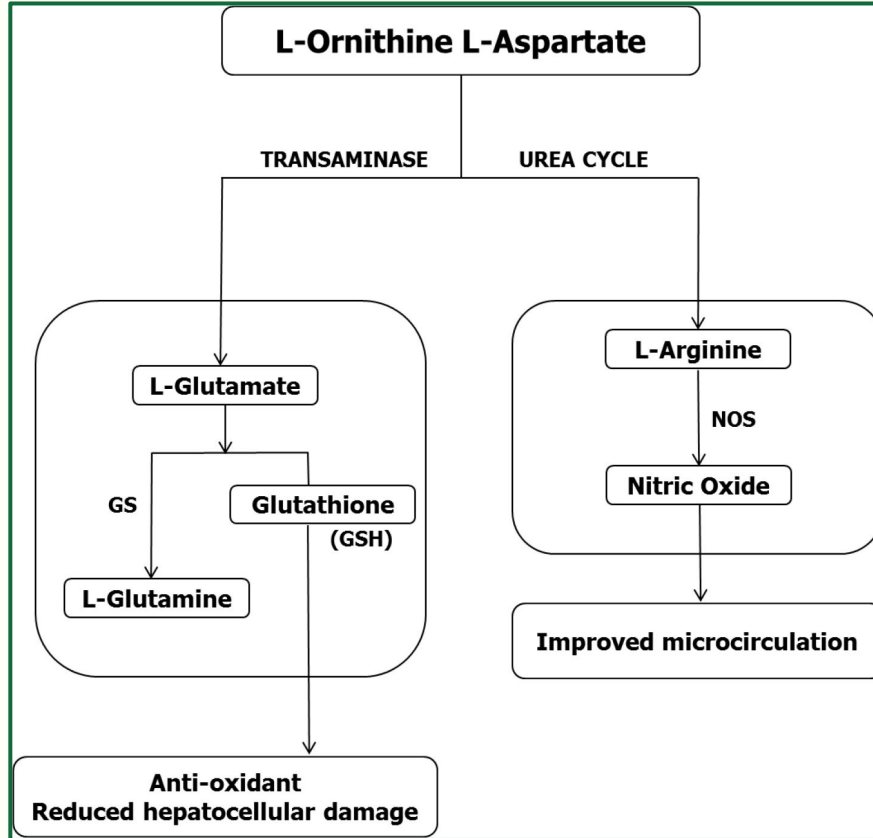


Terapi LOLA memutus siklus

Amonia ↓↓

Fungsi otot & otak pulih

Mekanisme Kerja LOLA



03

Journal Support

LOLA pada MHE & Pencegahan OHE

Systematic Review

Efficacy of L-ornithine L-aspartate for minimal hepatic encephalopathy in patients with cirrhosis: A meta-analysis of randomized controlled trials

Qiufeng He¹, Chuangjie Mao¹, Zhili Chen¹, Yilan Zeng, Yang Deng, Rong Hu^{*}

Department of Hepatology, Public Health Clinical Center of Chengdu, NO.377, Jing Ming Road, Jin Jiang District, Chengdu, Sichuan, China

Metode:

Meta-analisis 6 RCT (292 pasien minimal HE) menilai LOLA vs plasebo

Intervensi:

LOLA oral/intravena berbagai dosis (15-20G/hari), durasi bervariasi (7-90 hari)

Hasil:

LOLA dua kali lebih efektif membalik MHE (RR 2,264) daripada plasebo

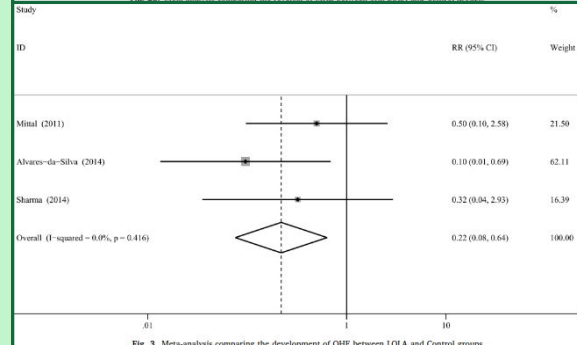
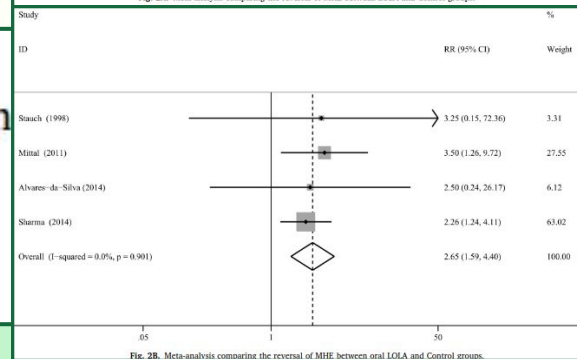
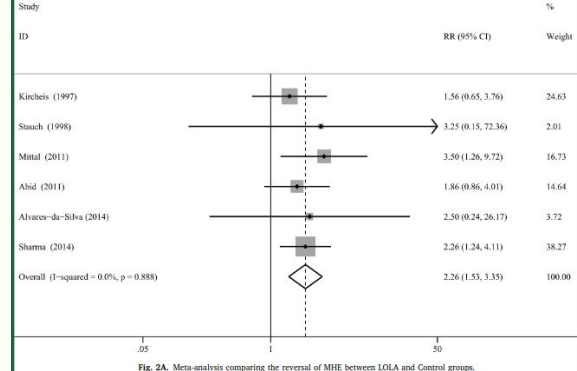
Risiko progresi ke OHE menurun (RR 0,22)

LOLA oral lebih efektif daripada LOLA intravena (RR 2,648 vs 1,669)

Tidak ada perbedaan mortalitas atau kadar amonia akhir

Kesimpulan:

LOLA efektif untuk MHE dan pencegahan OHE



Beneficial effects of L-ornithine L-aspartate for prevention of overt hepatic encephalopathy in patients with cirrhosis: a systematic review with meta-analysis

Roger F. Butterworth¹ 

Received: 14 March 2019 / Accepted: 8 July 2019 / Published online: 23 July 2019

LOLA pada Pencegahan OHE

Metode:

Meta-analisis 6 RCT (384 pasien sirosis) LOLA untuk pencegahan progresi OHE

Intervensi:

LOLA oral/intravena dosis variasi 6–30g/hari, 7–90 hari vs plasebo

Hasil:

LOLA mengurangi progresi minimal HE ke overt HE / OHE (RR 0,23)

LOLA juga efektif pasca perdarahan varises (RR 0,42) & post-TIPS (RR 0,30)

Baik LOLA oral maupun IV menurunkan kadar amonia

Kesimpulan:

LOLA efektif untuk pencegahan OHE

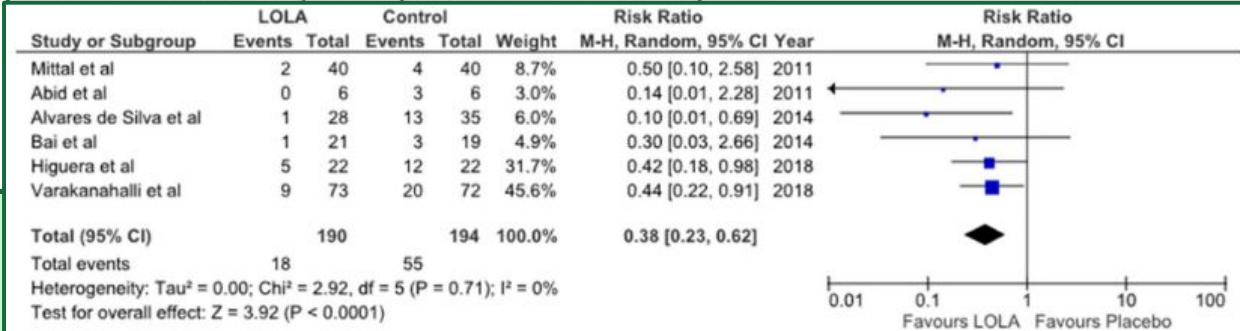


Fig. 1 Forest Plot for the efficacy of LOLA versus placebo/no intervention for the prevention of progression of MHE to OHE (Abid et al. 2011; Mittal et al. 2011; Alvares-da-Silva et al. 2014), secondary OHE prophylaxis (Varakanahalli et al. 2018), primary OHE prophylaxis

(Higuera-de-la-Tijera et al. 2018) or post-TIPSS OHE prophylaxis (Bai et al. 2014) from results of the appropriate published RCTs. Trials are identified by first author names, number of the reference in parentheses and year of publication



Fig. 2 Forest plot for the efficacy of LOLA versus placebo/no intervention for the prevention of progression from MHE to OHE in patients with cirrhosis from results of published RCTs. Trial identification as for legend to Fig. 1

Efficacy of L-Ornithine L-Aspartate for the Treatment of Hepatic Encephalopathy and Hyperammonemia in Cirrhosis: Systematic Review and Meta-Analysis of Randomized Controlled Trials

Roger F. Butterworth^{*}, Gerald Kircheis[†], Norbert Hilger[‡], Mark J.W. McPhail[§]

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Efikasi LOLA pada Status Mental & Amonia HE

Metode:

Meta-analisis 10 RCT (884 pasien) membandingkan LOLA dengan plasebo/standar terapi

Intervensi:

LO oral/intravena durasi 7–30 hari, variasi dosis 15–20g/haru

Hasil:

LO meningkatkan status mental (RR 1,36):

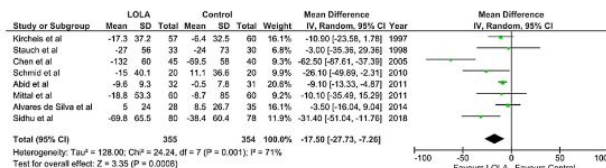
Efek kuat pada MHE (RR 2,15), dan efek moderat pada OHE (RR 1,19)

LO menurunkan amonia sebesar –17,5 $\mu\text{mol/L}$

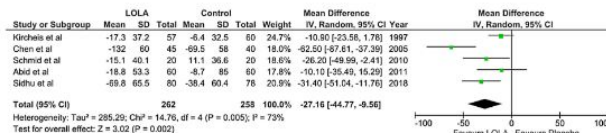
Kesimpulan:

LOLA efektif menurunkan amonia dan memperbaiki status mental pasien HE

A. Ammonia lowering–all studies



B. Ammonia lowering–IV LOLA



C. Ammonia lowering–PO LOLA

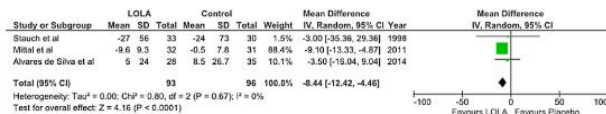
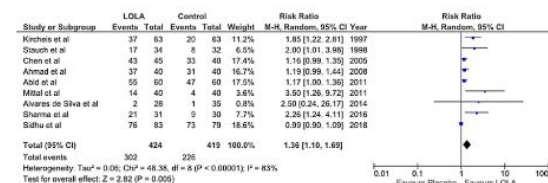
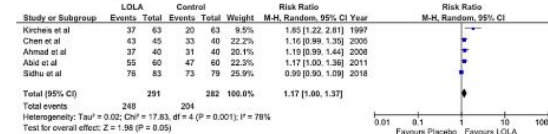


Figure 3 Forest plots indicating the pooled effect of LOLA versus placebo/no intervention controls for the lowering of blood ammonia in (A) all patients in trials selected according to inclusion/exclusion criteria irrespective of the type of LOLA formulation, (B) and (C) patients selected according to the LOLA formulation (intravenous, oral). LOLA: L-Ornithine L-Aspartate, RR: Risk Ratio, CI: Confidence Interval.

A. Mental state improvement–all studies



B. Mental state improvement–IV LOLA



C. Mental state improvement–PO LOLA

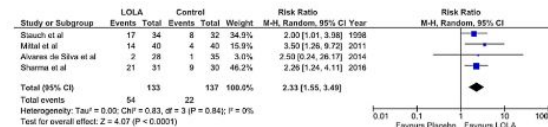


Figure 4 Forest plots indicating the pooled effect of LOLA versus placebo/no intervention controls for the improvement in mental state for all forms of HE in (A) all patients in trials selected according to inclusion/exclusion criteria irrespective of the type of LOLA formulation, (B) and (C) patients selected according to the LOLA formulation (intravenous, oral). LOLA: L-Ornithine L-Aspartate, RR: Risk Ratio, CI: Confidence Interval, HE: Hepatic Encephalopathy.

LOLA Oral vs Intravenous

Intravenous versus oral 'L-ornithine-L-aspartate' in overt hepatic encephalopathy: a randomized comparative study

Ashok Jhajharia^{1,2}, Shashank Singh¹, Sangeeta Jana¹, Prachis Ashdhir¹ & Sandeep Nijhawan¹

Metode:
RCT 40 pasien OHE
Intervensi:
LOLA oral 5g tiap 4 jam vs LOLA IV 30 g/hari, 5 hari
Hasil:
Amonia menurun signifikan di kedua kelompok
Perbedaan antar kelompok tidak signifikan
Kesimpulan:
LOLA oral maupun IV efektif menurunkan kadar amonia

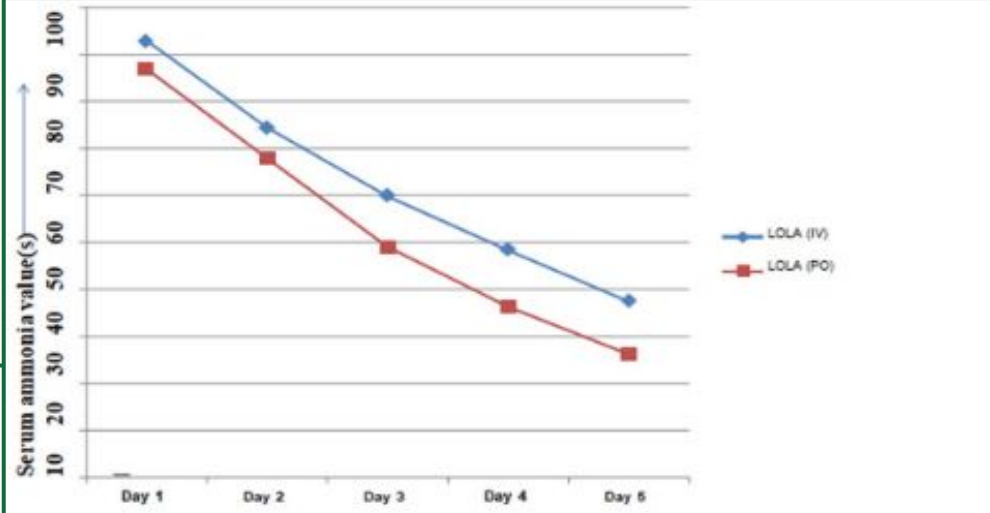


Figure 2. Line graph: day-wise linearity of serum ammonia level in both groups.

	LOLA (IV)	LOLA (ORAL)	P-value (95% C.I.)
Day 1	92.65 ± 28.81	86.75 ± 19.11	0.450
Day 2	74.2 ± 29.89	67.8 ± 20.8	0.437
Day 3	59.6 ± 25.29	48.85 ± 19.93	0.144
Day 4	48.2 ± 26.03	36.1 ± 15.52	0.082
Day 5	37.25 ± 23.08	26 ± 13.12	0.066
ΔNH ₃ (day 1-Day 5)	55.4 ± 32.58	60.75 ± 13.82	0.511
ΔNH ₃ (within group)	<0.001 (S)	<0.001 (S)	

Table 2. Serum ammonia (NH₃) levels among study groups.

Efficacy and safety of L-ornithine L-aspartate combined with lactulose in treatment of hepatic encephalopathy: a systematic review and meta-analysis of randomized controlled trial

Hui Zhang¹, Yujuan Fu¹, Minghao Lin¹, Zheng Nan^{1,2} and Dexi Zhao^{1,2*}

¹Changchun University of Chinese Medicine, Changchun, China, ²The Affiliated Hospital to Changchun University of Chinese Medicine, Changchun, China

LOLA + Laktulosa
Metode:
Meta-analisis 12 RCT melibatkan 858 pasien HE
Intervensi:
Injeksi LOLA 10 g/hari + laktulosa vs laktulosa saja
Hasil
Total efektivitas meningkat 31 % dibanding laktulosa saja (RR 1,31; P<0,00001)
AST, ALT, bilirubin & amonia turun signifikan
Kombinasi aman; tidak meningkatkan efek samping
Kesimpulan:
LOLA + laktulosa lebih efektif dibanding laktulosa sendiri

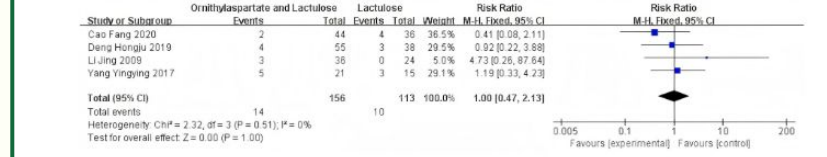


FIGURE 6
Forest plot of adverse reactions.

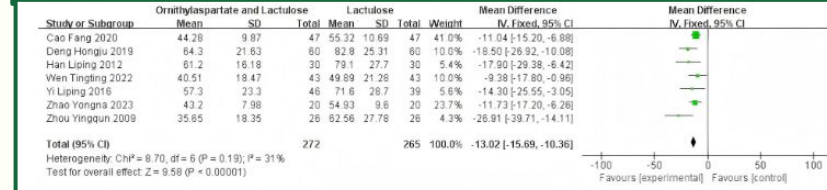


FIGURE 7
Forest plot of NH3.

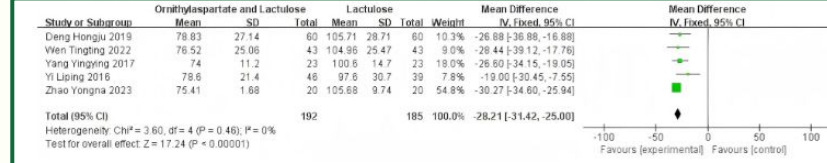


FIGURE 8
Forest plot of AST.

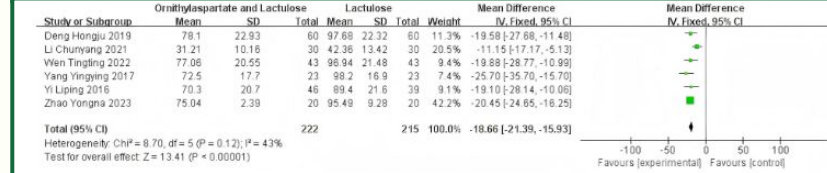


FIGURE 9
Forest plot of ALT.

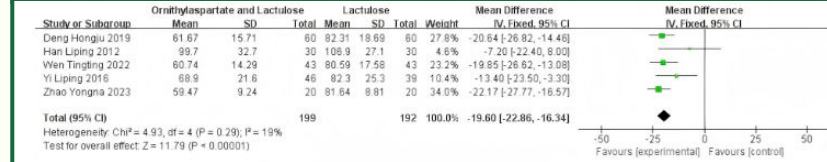


FIGURE 10
Forest plot of TBIL.

Take Home Messages

CLD global: >2 juta kematian per tahun

HE terjadi oleh hiperamonemia dari gangguan siklus urea dan shunt portal

LOLA → aktifkan CPS1 & transaminasi → percepat detoks amonia

LOLA baik oral maupun intravena konsisten menurunkan amonia & memperbaiki status mental pada HE, dengan kombinasi LOLA + laktulosa lebih unggul daripada laktulosa saja

LOLA aman dan tolerabel, membantu mencegah progresi HE